

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Technology, Grade 11

Workplace Preparation

TMJ3E

This hands-on, project-based course is designed for students planning to enter an occupation or apprenticeship in manufacturing directly after graduation. Students will work on a variety of manufacturing projects, developing knowledge and skills in design, fabrication, and problem solving and using tools and equipment such as engine lathes, milling machines, and welding machines. In addition, students may have the opportunity to acquire industry-standard certification and training. Students will develop an awareness of environmental and societal issues related to manufacturing and will learn about secondary school pathways that lead to careers in the industry.

Prerequisite: None

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the primary manufacturing industries and the processes and technologies related to them;
- A2.** demonstrate an understanding of how a design process is used in the planning and development of a manufacturing project;
- A3.** describe and demonstrate the correct use of processes required for making material conversions;
- A4.** demonstrate an understanding of relevant mathematical and scientific concepts and apply technological literacy and communication skills in the study of manufacturing technology.

SPECIFIC EXPECTATIONS

A1. Primary Manufacturing Industries

By the end of this course, students will:

- A1.1** describe how materials from primary manufacturing industries (*e.g., steel, plastics, lumber, paper*) are used in the manufacturing industry;
- A1.2** identify and describe the key areas of operation in primary manufacturing industries (*e.g., design, planning, fabrication, inspection, worker safety*);
- A1.3** describe the influence that primary manufacturing industries have on secondary manufacturing industries (*e.g., in terms of availability of materials, cost, viability*).

A2. Design Process

By the end of this course, students will:

- A2.1** describe the steps of a design process used in the planning and development of a manufacturing project (*e.g., brainstorming, determining order of operations, developing flow charts, conducting a needs analysis, creating mock-ups*) (see pp. 22–23);
- A2.2** describe how various forms of communication (*e.g., group discussion, brainstorming*), graphic representation (*e.g., sketches, technical drawings, computer-aided design [CAD]*), and documentation (*e.g., research reports*) are used in the development of product ideas;

- A2.3** identify and describe skills and attitudes needed to work as a cooperative member and/or leader of a group during the design process (*e.g., effective time management, goal setting, communication skills, respect for differences, ability to delegate tasks*);

- A2.4** explain why technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) are important considerations in the design process (see pp. 7–8).

A3. Material Conversion

By the end of this course, students will:

- A3.1** describe and demonstrate the correct use of a variety of processes for joining materials (*e.g., shielded metal arc welding [SMAW], gas metal arc welding [GMAW], brazing, soldering, riveting, spot welding, crimping, bonding*);
- A3.2** describe and demonstrate the correct use of a variety of processes for cutting or separating materials (*e.g., using oxy-fuel torches, saws, mills, lathes, shears, plasma cutters*);
- A3.3** describe and demonstrate the correct use of a variety of processes for forming materials, using various tools and equipment (*e.g., hammers, box and pan brake, benders, rollers*);
- A3.4** demonstrate a working knowledge of basic metallurgy (*e.g., flame hardening, case hardening, tempering, annealing*).

A4. Mathematics, Science, and Technological Literacy

By the end of this course, students will:

- A4.1** demonstrate a working knowledge of various mathematical formulas (*e.g., related to trigonometry, geometry, force/power*) and applications (*e.g., tap drill sizing, calculating machine speeds and feeds, indexing*) that are commonly used in manufacturing;
- A4.2** use mathematical calculations to solve manufacturing problems (*e.g., use algebraic equations; calculate perimeters, volumes, areas*), using appropriate resources (*e.g., tables, charts*);
- A4.3** apply mathematical concepts (*e.g., linear systems, angle calculations, integers, orders of operations, decimals and fractions*) and skills in performing a variety of tasks required within the context of manufacturing design and production (*e.g., percent/decimal/fraction conversions, US customary/British imperial and metric unit conversions*);
- A4.4** identify the properties (*e.g., magnetic, physical, mechanical, thermal, electrical*) of natural and manufactured solid materials;
- A4.5** describe how the physical properties of solid materials (*e.g., density, appearance, porosity, surface texture*) create advantages or disadvantages that affect their suitability for use in manufacturing;
- A4.6** describe how the mechanical properties of solid materials (*e.g., hardness, ductility, elasticity, brittleness, compression, fatigue, shear, tensility*) create advantages or disadvantages that affect their suitability for use in manufacturing;
- A4.7** demonstrate effective use of technological literacy and communication skills to interpret and communicate information related to manufacturing technology (*e.g., in technical manuals, reports, presentations*).

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use technical design and production skills to interpret and prepare drawings and to develop process plans for a manufacturing project;
- B2.** demonstrate the ability to select appropriate materials for the manufacture of a product and prepare them properly for production;
- B3.** demonstrate a working knowledge of various metrology tools used to measure, lay out, and inspect a product;
- B4.** demonstrate a working knowledge of the purpose, characteristics, and safe use of various hand tools, machine tools, power tools, and equipment used in the manufacture of products.

SPECIFIC EXPECTATIONS

B1. Technical Drawings and Process Plans

By the end of this course, students will:

- B1.1** demonstrate the ability to accurately interpret engineering drawings (*e.g., with respect to dimensions, welding symbols, weldments, notes and specifications, tolerances*) related to the production of a project;
- B1.2** use appropriate sketching and/or drawing techniques to prepare working drawings (*e.g., sketches, scale drawings, perspectives, isometric and oblique views, assembly and detail drawings, pattern and layout development*) that convey ideas and processes and show materials and specifications;
- B1.3** develop a process plan for the production of a product based on design criteria and specifications (*e.g., create a work plan, identify appropriate tools and equipment, produce a bill of material, develop a budget for the project, develop a flow chart of the project*).

B2. Selection and Preparation of Materials

By the end of this course, students will:

- B2.1** select appropriate materials for the manufacture of a product based upon design criteria

(*e.g., criteria related to pattern and layout, cutting, squaring, drilling, machining, weld preparation and procedures*);

- B2.2** use the proper procedures to prepare materials for production (*e.g., measuring, marking, grinding, cutting*).

B3. Metrology Tools

By the end of this course, students will:

- B3.1** select and use a variety of measuring tools (*e.g., calipers, scales, micrometers, vernier calipers, dial indicators*) to measure projects according to specifications;
- B3.2** demonstrate the use of precise layout techniques for projects using the proper equipment (*e.g., layout dye, scribe, marking punch, precision straight edge and square, vernier height gauge*);
- B3.3** use inspection techniques (*e.g., inspection reports, destructive and non-destructive testing techniques, rubrics*) to evaluate projects and ensure quality;
- B3.4** apply principles of dimensional metrology (*e.g., precision measurement, tolerancing for interchangeable manufacturing*) to manufacturing processes.

B4. Selection and Use of Tools and Equipment

By the end of this course, students will:

- B4.1** demonstrate the correct selection and safe and proper use of hand and power tools and/or equipment (*e.g., lathe, mill, drill press, wrenches, tin snips, shears, rivet guns, foot and hand brakes, rollers, bar folds, punching and notching machines; equipment for shielded metal arc welding [SMAW], gas metal arc welding [GMAW], and gas tungsten arc welding [GTAW]; spot welder, plasma cutter, oxy-fuel torch, soldering gun*) when manufacturing a product;
- B4.2** use the appropriate tools and/or equipment (*e.g., fasteners, rivets, tape, self-tapping screws*) for various assembly techniques (*e.g., welding, gluing/bonding*);
- B4.3** demonstrate a working knowledge of preventive maintenance (*e.g., cleaning, lubricating, sharpening*) and proper storage of tools and equipment commonly used in manufacturing.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of ways in which the manufacturing industry affects the environment;
- C2.** explain how the manufacturing industry affects various aspects of society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** explain the importance of the proper storage, disposal, and recycling of obsolete and waste products in manufacturing;
- C1.2** describe the benefits of using environmentally friendly products in the manufacturing process (*e.g., the benefits of water-based versus solvent-based chemicals*);
- C1.3** explain how various sources of power generation used in manufacturing (*e.g., coal, nuclear, solar, wind, hydrogen fuel cell, tidal, geothermal*) affect the environment;
- C1.4** identify conservation strategies that the manufacturing industry could employ (*e.g., minimize water usage, convert to energy-efficient lighting, exploit transportation efficiencies, reduce paper usage by communicating electronically*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** explain how the manufacturing industry affects the economy of their community or region;
- C2.2** explain how manufacturing affects people's daily lives;
- C2.3** identify ways in which the manufacturing industry affects the culture and society of a community or region (*e.g., by creating employment opportunities; supporting social and cultural activities in the community; increasing industrial activity in the community/region, which some people may see as a threat to their way of life and/or the environment*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with health and safety legislation, standards, and practices as they relate to processes, materials, tools, and equipment used in manufacturing;
- D2.** demonstrate an understanding of career opportunities in the manufacturing industry and the education, training, and certification required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and explain the importance of the specific components of legislation and standards related to workplace safety in the manufacturing industry (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (e.g., *Workplace Safety and Insurance Board [WSIB]*, *Industrial Accident Prevention Association [IAPA]*, *Construction Safety Association of Ontario [CSAO]*);
- D1.2** demonstrate good housekeeping practices in the work environment (e.g., *preventing and cleaning up spills and leaks, keeping areas clean and clear of obstructions, properly organizing tools and equipment*);
- D1.3** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);
- D1.4** describe health and safety roles, responsibilities, and procedures in manufacturing (e.g., *concerning choice of equipment and materials, maintenance of equipment, storing of materials and equipment, inspection of facilities and equipment, in-service and training*);
- D1.5** demonstrate the safe use of tools and equipment in compliance with safety manuals, instructions, and institutional requirements;

- D1.6** use protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify employment opportunities available in the manufacturing industry locally, provincially, and nationally;
- D2.2** identify the training and certification required (e.g., *by the Ministry of Training, Colleges and Universities – Workplace Training Branch*) to become a skilled tradesperson in the manufacturing industry;
- D2.3** research and report on opportunities available locally to obtain certifications and/or training that it would be beneficial to have if they are pursuing a career in manufacturing (e.g., *cardio-pulmonary resuscitation [CPR]*, *Standard First Aid*, *fall protection training*, *Canadian Welding Bureau certification*);
- D2.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the manufacturing industry (e.g., *mentoring programs*, *virtual networking/support groups*, *specialized postsecondary programs*, *relevant trade/industry associations*);
- D2.5** demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (e.g., *decision making*, *job task planning and organizing*, *problem solving*);

D2.6 demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, initiative, working independently*);

D2.7 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (*e.g., Passport to Safety certificate, Ontario Skills Passport Work Plan, photographs of a project, technical drawings, reports, assignments*), and explain why having a current portfolio is important for career development and advancement.